Chemistry Topic 8 — Nuclear Chemistry, Explosives and Environment

Sources baked in: NCERT Class 9, 11 & 12 (Nuclear Chemistry; Environmental Chemistry; Radioactivity), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Recurring questions on Radioactivity as a nuclear property, Hydrogen bomb based on fusion (not fission), Dynamite invented by Alfred Nobel (nitroglycerin), RDX chemical identity, Argon not being a greenhouse gas, Acid rain caused by SO_2 and NO_2, and Minamata disease by Mercury. **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**Global Carbon & Methane Pledges**	Enhanced monitoring of non-CO_2 greenhouse gases under COP agreements. Methane (CH_4) possesses a **Global Warming Potential (GWP)** $\approx 28–36$ **times greater than** CO_2 over a 100-year timescale, primarily emanating from flooded paddy cultivation, livestock rumination, and coalbed degassing.
2025	**UPPCS Prelims 2025**	Direct question on **Stratospheric Ozone Layer & CFCs**: Both statements confirmed correct—the ozone layer shields the biosphere from lethal solar UV-B rays, and chlorofluorocarbons (CFCs) trigger catalytic ozone destruction via free chlorine radicals ($Cl^{\bullet}$).
2024	**UPPCS Prelims 2024**	Assertion-Reason on **Acid Rain Chemistry**: Rainwater acidity is driven by atmospheric SO_2 and NO_x dissolving to form sulphuric (H_2SO_4) and nitric (HNO_3) acids, with pH dropping below 5.6. Causes severe "stone cancer" on white marble monuments ($CaCO_3 \rightarrow CaSO_4$).
2022	**UPPCS Prelims 2022**	Greenhouse Gas Identification: **Argon (Ar)** is strictly **NOT** a greenhouse gas. Diatomic/monatomic symmetric gases (N_2, O_2, Ar) lack dipole moments during vibrational excitation and cannot absorb infrared radiation.
2021	**UPPCS Prelims 2021**	Nuclear vs Atomic Trap: **Radioactivity is strictly a nuclear property** (originating in the unstable nucleus), completely unaffected by chemical bonds, temperature, or pressure. Confirmed the **Hydrogen Bomb operates on Nuclear Fusion**, using a fission device only as a detonating trigger.

Detailed Theory Notes

8.1 Radioactivity Kinetics & Radiometric Dating Principles

1. Fundamental Decay Physics

- Unstable nuclei with atomic numbers $Z > 82$ (beyond Lead) possess excess positive charges causing electrostatic repulsion that overcomes the attractive strong nuclear force.
- The decay rate is purely statistical:

$$\frac{dN}{dt} = -\lambda N$$

  Integration yields the exponential law: $N(t) = N_0 e^{-\lambda t}$.
- **Activity (A)**: The rate of nuclear disintegrations per unit time ($A = \lambda N$). Measured in Becquerels (Bq).

2. Radiocarbon Dating (^{14}C)

- Natural nitrogen in the upper atmosphere is bombarded by cosmic-ray neutrons:

$$^{14}_{7}N + ^{1}_{0}n \rightarrow ^{14}_{6}C + ^{1}_{1}p$$

- ^{14}C rapidly oxidizes into $^{14}CO_2$, which enters the terrestrial carbon cycle via plant photosynthesis and food chains.
- While an organism is alive, continuous metabolic exchange maintains a constant equilibrium ratio of $^{14}C/^{12}C \approx 1 : 10^{12}$.
- Upon death, metabolic assimilation ceases. The radioactive ^{14}C within tissues disintegrates with a half-life of $5,730 \text{ years}$:

$$^{14}_{6}C \xrightarrow{\beta^-} ^{14}_{7}N + ^{0}_{-1}e$$

- Measuring the remaining $^{14}C/^{12}C$ ratio via accelerator mass spectrometry enables precise dating of organic relics up to $\approx 50,000 \text{ years}$.

8.2 Nuclear Weapons & Reactor Core Physics

1. Fission vs. Fusion Energy Dynamics

- **Nuclear Fission**: Heavy nuclei ($A \approx 235$) have a binding energy per nucleon of $\approx 7.6 \text{ MeV}$. Splitting into fragments ($A \approx 120$) with higher binding energy ($\approx 8.5 \text{ MeV}$) releases $\approx 0.9 \text{ MeV}$ per nucleon, or $\approx 200 \text{ MeV}$ per fission event.
- **Nuclear Fusion**: Light nuclei ($^2H, ^3H$) with low binding energy ($\approx 1.1 \text{ to } 2.8 \text{ MeV}$) fuse into tightly bound ^4He ($\approx 7.07 \text{ MeV/nucleon}$), releasing $\approx 17.6 \text{ MeV}$ for only 5 nucleons, yielding $\approx 3.5 \text{ MeV}$ per nucleon. Thus, **gravimetrically, nuclear fusion yields roughly four times more energy per kilogram of fuel than nuclear fission**.

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[ Nuclear Fission ]                          [ Nuclear Fusion ]
Splits heavy nuclei (U-235, Pu-239)          Fuses light nuclei (Deuterium + Tritium)
Triggered by slow thermal neutrons            Triggered by extreme heat (>10⁷ K) & pressure
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Used in Nuclear Reactors & Atomic Bombs        Powers the Sun & Hydrogen Bombs
High radioactive fission waste products        Minimal direct radioactive waste products
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8.3 Atmospheric Chemistry: Acid Rain & Ozone Depletion

1. Acid Rain Dynamics

- High-temperature combustion in automobile engines oxidizes atmospheric nitrogen to nitric oxide:

$$N_2 + O_2 \xrightarrow{>1500^\circ C} 2NO, \quad 2NO + O_2 \rightarrow 2NO_2$$

- Coal combustion oxidizes pyrite and organic sulphur impurities:

$$S + O_2 \rightarrow SO_2$$

- In the presence of suspended particulate matter (soot, metal dust) and sunlight, SO_2 and NO_2 undergo aqueous-phase catalytic oxidation in cloud droplets into H_2SO_4 and HNO_3.
- Marble stone corrosion (Stone Leprosy):

$$CaCO_3\ (\text{Insoluble marble}) + H_2SO_4 \rightarrow CaSO_4 \cdot 2H_2O\ (\text{Soluble gypsum}) + CO_2 \uparrow$$

2. Stratospheric Ozone Destruction by CFCs

- The stability of CFCs enables them to survive intact in the troposphere for 50–100 years.
- In the stratosphere, solar UV radiation photolyzes CFCs to release atomic chlorine:

$$\text{Initiation: } CF_2Cl_2 \xrightarrow{h\nu\ (\lambda<220\ \text{nm})} CF_2Cl^\bullet + Cl^\bullet$$

$$\text{Propagation 1: } Cl^\bullet + O_3 \rightarrow ClO^\bullet + O_2$$

$$\text{Propagation 2: } ClO^\bullet + O^\bullet \rightarrow Cl^\bullet + O_2$$

$$\text{Net Reaction: } O_3 + O^\bullet \rightarrow 2O_2$$

- The $Cl^\bullet$ radical acts as a true homogeneous catalyst: it accelerates the destruction of ozone without being consumed in the net reaction.

Bilingual Terminology -- Radiation, Nuclear and Environmental Chemistry

English Term	Hindi Term	Key Anchor
Radiation (विकिरण)	विकिरण	Energy emitted as electromagnetic waves or particles
Ionising Radiation (आयनकारी विकिरण)	आयनकारी विकिरण	High-energy radiation that removes electrons from atoms; alpha, beta, gamma, X-ray
Non-ionising Radiation (अआयनकारी विकिरण)	अनायनकारी विकिरण	Lower energy; radio waves, microwaves, visible light, UV-A
Radioactive Waste (रेडियोधर्मी अपशिष्ट)	रेडियोधर्मी अपशिष्ट	Waste from nuclear reactors, hospitals; requires careful disposal

English Term	Hindi Term	Key Anchor
Half-life (अर्ध-जीवन)	अर्ध-जीवन	Time for half of radioactive atoms to decay
Radiocarbon Dating (रेडियोकार्बन काल-निर्धारण)	रेडियोकार्बन डेटिंग	Uses C-14 decay to date organic material up to ~50,000 years
UV Radiation (पराबैंगनी विकिरण)	पराबैंगनी विकिरण	Electromagnetic radiation 10-400 nm; UV-B causes sunburn; blocked by ozone
Infrared Radiation (अवरक्त विकिरण)	अवरक्त / ताप विकिरण	Electromagnetic radiation 700 nm-1 mm; heat radiation
Microwave (सूक्ष्म तरंग)	सूक्ष्म तरंग	1 mm-1 m wavelength; RADAR; microwave oven (2.45 GHz)
Air Pollution (वायु प्रदूषण)	वायु प्रदूषण	Contamination of air with harmful gases/particles
Water Pollution (जल प्रदूषण)	जल प्रदूषण	Contamination of water bodies
Acid Rain (अम्ल वर्षा)	अम्लीय वर्षा	pH < 5.6; H2SO4 + HNO3 from SO2 + NOx emissions
Eutrophication (सुपोषण)	पोषकता वृद्धि	Excess nutrients in water -> algae bloom -> O2 depletion
BOD (जैव रासायनिक ऑक्सीजन मांग)	जैव रासायनिक ऑक्सीजन माँग	Biochemical Oxygen Demand; measures organic water pollution
Photochemical Smog (प्रकाश रासायनिक धुंध)	प्रकाश-रासायनिक स्मॉग	NO + VOC + sunlight -> ozone + PAN; brown smog; Los Angeles type
e-Waste (इलेक्ट्रॉनिक कचरा)	इलेक्ट्रॉनिक कचरा	Discarded electronic devices; contains toxic metals (Pb, Cd, Hg)

One-Liner Revision -- Radiation, Nuclear and Environmental Chemistry

#	Fast Fact
1	Ionising radiation: alpha, beta, gamma, X-rays, cosmic rays -- can cause cancer
2	Non-ionising: radio, microwave, visible, UV-A -- generally safer (UV-B intermediate)
3	UV-B (280-315 nm) = causes sunburn, skin cancer; blocked by ozone layer
4	UV-A (315-400 nm) = causes tanning; glass blocks UV; used in sterilisation
5	Microwave oven = 2.45 GHz; water molecules absorb and vibrate -> heat food
6	C-14 half-life = **5730 years**; used for dating organic material
7	Radioactive decay: alpha (weakest penetration), beta, gamma (strongest penetration)
8	Radiation doses: mSv (millisievert); background dose ~2-3 mSv/year
9	Radiation therapy = high-dose gamma/X-ray to kill cancer cells
10	Chernobyl (1986) = RBMK reactor explosion; released Cs-137 and I-131

#	Fast Fact
11	Acid rain: SO2 + H2O -> H2SO3; SO3 + H2O -> H2SO4; causes damage to Taj Mahal marble
12	Acid rain pH < 5.6 (normal rain = 5.6 due to natural CO2)
13	London smog = industrial smog (SO2 + soot + fog); cold; 1952 Great Smog killed 4,000+
14	LA smog = photochemical smog (NOx + VOC + sunlight -> O3 + PAN); warm, sunny
15	Ozone (O3) in troposphere = pollutant (causes respiratory problems); in stratosphere = protective
16	PM2.5 = particulate matter < 2.5 micron; deepest lung penetration; most harmful
17	PM10 = particles < 10 micron; cause respiratory problems
18	BOD: high BOD = more organic pollution; clean water BOD < 3 mg/L
19	COD (Chemical Oxygen Demand) = total oxygen needed to chemically oxidise all organics
20	Eutrophication: excess N and P (from fertiliser runoff) -> algal bloom -> dead zone
21	Biomagnification = persistent pollutants (DDT, Hg, PCBs) concentrate up the food chain
22	Mercury (Hg) poisoning = Minamata disease (Japan, 1950s); methylmercury in fish
23	Cadmium poisoning = Itai-itai disease (Japan); bone pain; from mine runoff
24	Lead (Pb) poisoning = cognitive damage in children; banned from petrol (TEL) and paint
25	Arsenic contamination = groundwater in Bangladesh, West Bengal; skin lesions, cancer
26	Fluorosis = excess fluoride (>1.5 ppm) = mottled teeth (dental); skeletal damage (skeletal)
27	e-waste = toxic heavy metals; India = 3rd largest e-waste generator globally
28	Basel Convention (1989) = controls transboundary movement of hazardous wastes
29	Stockholm Convention (2001) = bans persistent organic pollutants (POPs) like DDT, PCBs
30	CPCB = Central Pollution Control Board (India); SPCB = State Pollution Control Board